Computer Organization & Architecture (BE04000251)

Unit-5
Microprogrammed Control Organization

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Outline

- Control Memory
- Address sequencing
- Micro program example
- Microinstruction Format

Control Memory

Control Memory

- Generally Computer contains **main 3 components**:

1. **CPU:** ALU(to perform computations), Registers(temporary storage of information), Control Unit(to generate control signals)
2. **I/O Devices**
3. **Memory**

❖ **<u>Use of Control Signals:</u>**

- By using control signals data will be transferred between registers.
- Control Unit instructs the ALU to perform the corresponding operations.
- It is the responsibility of the control unit to control the flow of program by generating the control signals.
- Examples: RD,WR etc.

Control Memory

- Two methods of implementing control unit are **hardwired control unit** and **microprogrammed control unit**.

Hardwired Control Unit	Microprogrammed Control Unit
Hardwired control unit generates the control signals **using logic circuits(Hardware).**	Microprogrammed control unit generates the control signals **with the help of micro instructions(Software) stored in control memory.**
Hardwired control unit is **faster** when compared to microprogrammed control unit as the required control signals are generated with the help of hardwares.	This is **slower** than the other as micro instructions are used for generating signals here.
Difficult to modify(done by designing) as the control signals that need to be generated are hard wired.	**Easy to modify(done by reprogramming)** as the modification need to be done only at the instruction level.

Control Memory

Hardwired Control Unit	Microprogrammed Control Unit
More costlier as everything has to be realized in terms of logic gates.	**Less costlier** than hardwired control as only micro instructions are used for generating control signals.
It **cannot handle complex instructions** as the circuit design for it becomes complex.	It **can handle complex instructions**.
Only **limited number of instructions are used(required large space for more instructions)** due to the hardware implementation.	Control signals **for many instructions(No more space is required as instructions are stored in memory)** can be generated.
Used in computer that makes use of **Reduced Instruction Set Computers(RISC).**	Used in computer that makes use of **Complex Instruction Set Computers(CISC).**

Control Memory

- A computer that contains microprogrammed control unit will have two separate memories: **main memory and control memory.**
- Main memory is available for user to store programs.
- Main memory consists of machine instructions and data.
- **Control memory can be a read only memory(ROM).**
- It holds a **fixed microprogram that cannot be alter.**
- Microprogram consists of microinstructions that specify various internal control signals for execution of microoperation.
- Each machine instructions initiates a series of microinstructions in control memory.
- These microinstructions generate the microoperations to fetch the instruction from memory, to evaluate the effective address, to execute the operation specified by instruction.
- The function of a control unit in a digital computer is to **initiate sequences of microoperations.**
- **Control words** can be programmed to perform various operations on the components of the system.
- A control unit whose binary control variables are stored in memory is called a **microprogrammed control unit**.

Control Memory

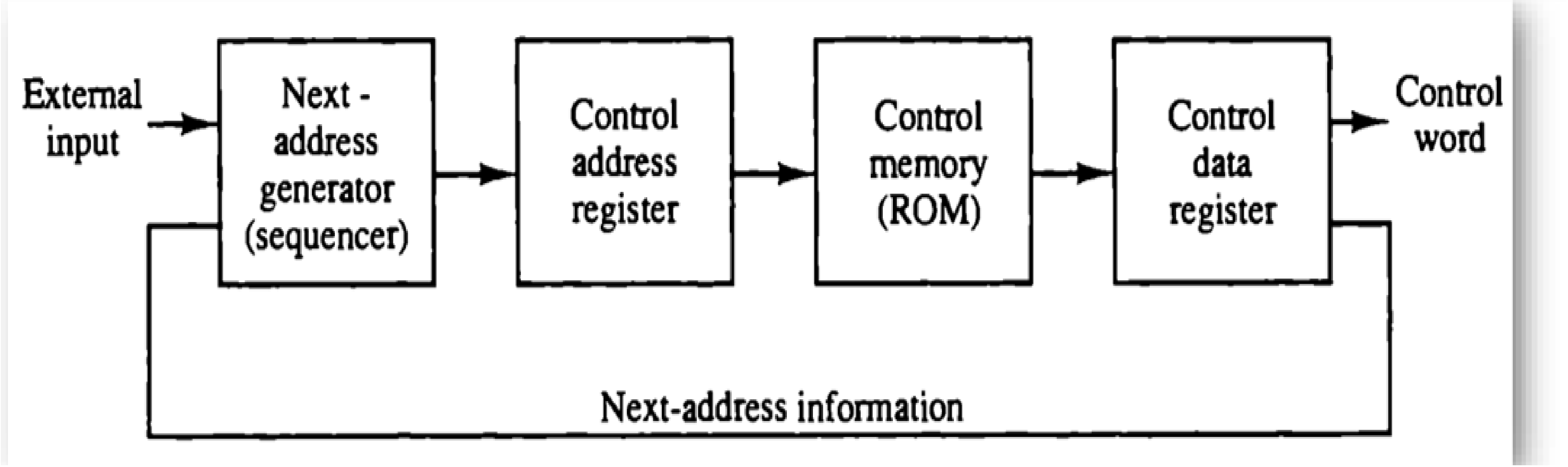
- Each word in control memory contains within it a **microinstruction**.
- The microinstruction specifies one or more microoperations for the system.
- A sequence of microinstructions is called **microprogrammed**.
- A memory that is part of a control unit is referred to as a **control memory.**

Microoperation	Microinstruction	Microprogram
Basic data manipulation	A control word that triggers microoperations	A sequence of microinstructions that define a machine instruction
PC ← PC + 1 (incrementing PC)	"Fetch instruction from memory" (MAR ← PC)	"Complete ADD instruction execution sequence"

Control Memory

- **Control Word:** The control variables at any given time can be represented by a string of 1's and 0's called a control word.
- **Control Memory:** It is the **storage in the microprogrammed control unit to store the microprogram.**
- **Microprogram:** Program stored in memory that generates all the control signals required to execute the instruction set correctly. It consists of **sequence of microinstructions(contains one or more microoperations)**.
- Since alterations of the microprogram are not needed once the control unit is in operation, the control memory can be a read-only memory (ROM). The content of the words in ROM are fixed and cannot be altered by simple programming since no writing capability is available in the ROM.
- **Microinstruction:** It specifies one or more micro operations for the system.

Microprogrammed Control organization



- Microprogrammed Control organization means the control signals are implemented with a program.
- Without address there is no meaning of memory.
- **CAR**: to hold adress of current instruction.
- **Control Memory**: to select instruction(control word/binary value control signal).
- **CDR**: holds the control word.
- **Sequencer**: holds the address of the next instruction.

Microprogrammed Control organization

- The control memory is assumed to be a **ROM**, within which all control information is permanently stored.
- The **control memory address register** specifies the address of the microinstruction, and the **control data register** holds the microinstruction read from memory.
- The microinstruction contains a **control word** that specifies one or more micro operations for the data processor. Once these operation are executed, the control must determine the next address.
- The location of the next microinstruction may be the one next in sequence, or it may be located somewhere else in the control memory.
- While the micro operations are being executed, the next address is computed in the **next address generator** circuit and then transferred into the control address register to read the next microinstruction.
- Thus a **microinstruction contains bits for initiating micro operations in the data processor part and bits that determine the address sequence for the control memory**.

Microprogrammed Control organization

- The next address generator is sometimes called a **microprogram sequencer**, as it determines the address sequence that is read from control memory.
- Typical **functions of a microprogram sequencer** are incrementing the control address register by one, loading into the control address register an address from control memory, transferring an external address, or loading an initial address to start the control operations.

Address Sequencing

Address Sequencing(Microprogram Sequencer)

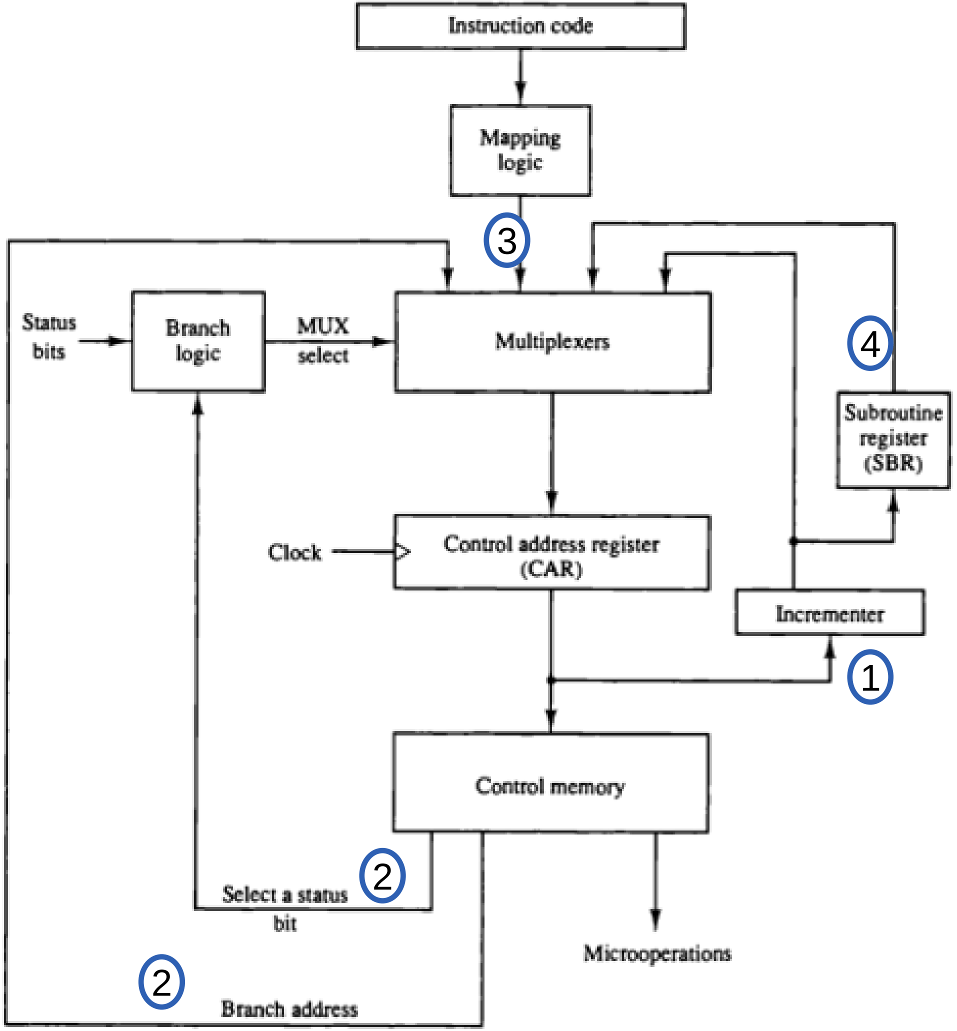
- Microinstructions are stored in control memory in groups, with each group specifynig a routine.
- Each computer instruction has its own microprogram routine in control memory to generate the microoperations that excute the instruction.
- Steps that the control must undergo during the execution of a single computer instruction.

1. Incrementing the content of control address register.
2. Unconditional branch or conditional branch, depending on status bit conditions.
3. A mapping process from the bits of the instruction to an address for control memory.
4. A facility for subroutine call and return.

GTU Question: Explain using a flowchart how address of control memory is selected in microprogrammed control unit.

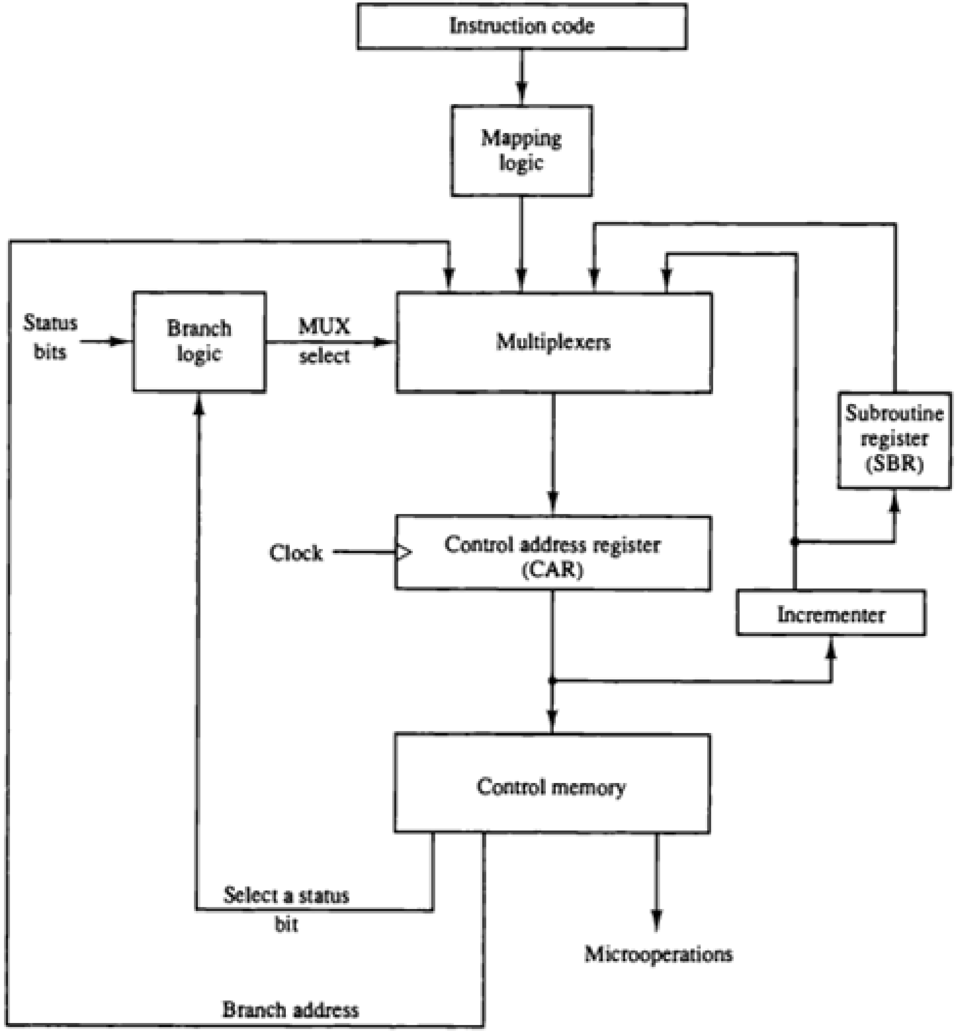
Selection of Next Microinstruction Address for Control Memory

- The microinstruction in control memory contains a **set of bits to initiate microoperations in computer registers and other bits to specify the method by which the next address is obtained.**
- The diagram shows four different paths from which the control address register (CAR) receives the address.
- The incrementer increments the content of the control address register by one, to select the next microinstruction in sequence.



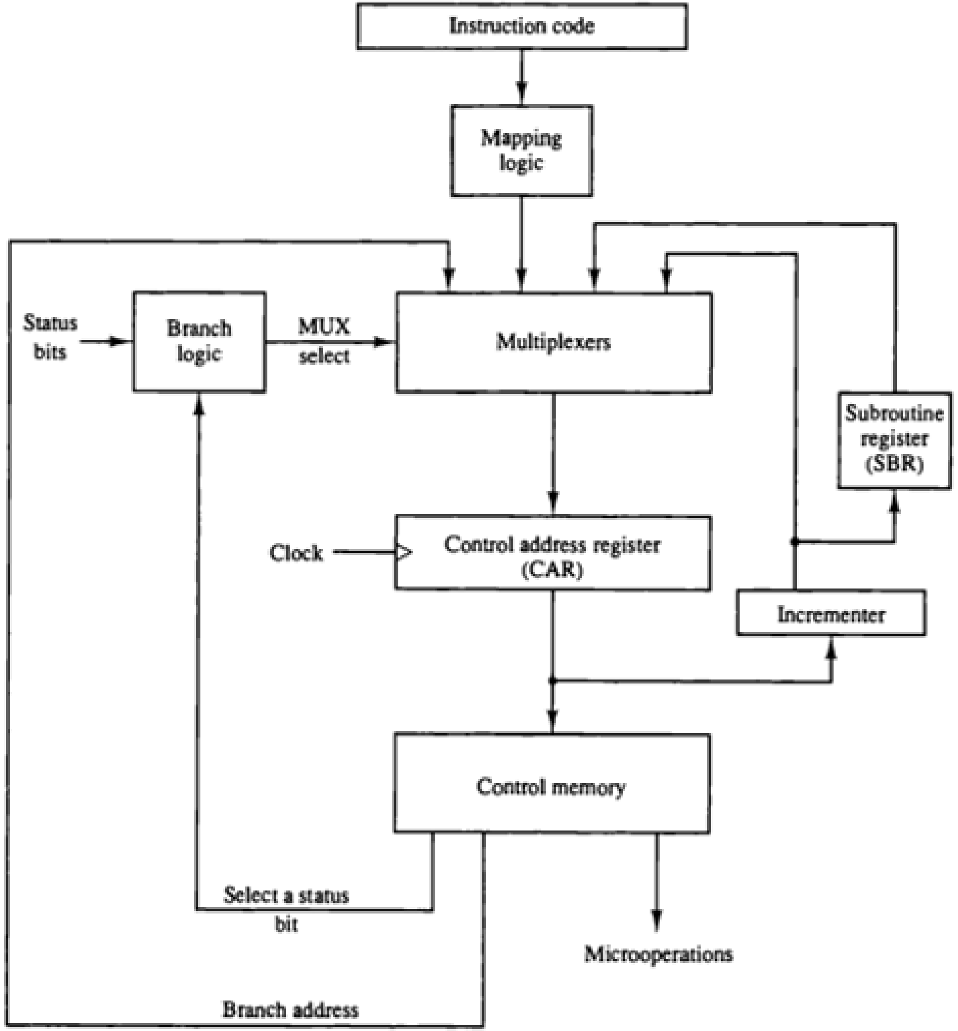
Selection of Address for Control Memory

- Branching is achieved by specifying the branch address in one of the fields of the microinstruction.
- Conditional branching is obtained by using part of the microinstruction to select a specific status bit in order to determine its condition.
- An external address is transferred into control memory via a mapping logic circuit.
- The return address for a subroutine is stored in a special register whose value is then used when the microprogram wishes to return from the subroutine.



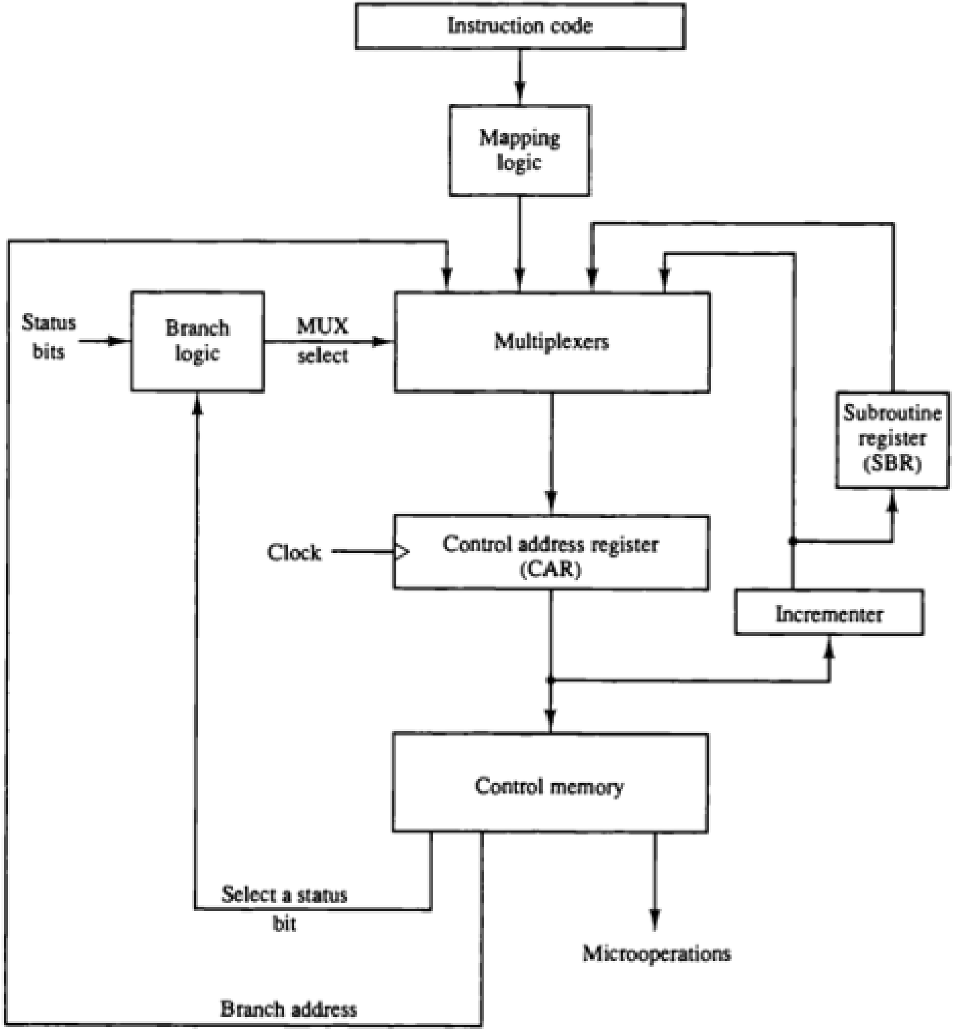
Selection of Address for Control Memory

- The status bits, together with the field in the microinstruction that specifies a branch address, control the conditional branch decisions generated in the branch logic.
- Suppose that there are **eight status bit conditions** in the system. Three bits in the microinstruction are used to specify any one of eight status bit conditions.
- These three bits provide the selection variables for the multiplexer. If the selected status bit is in the 1 state, the output of the multiplexer is 1 otherwise, it is 0.



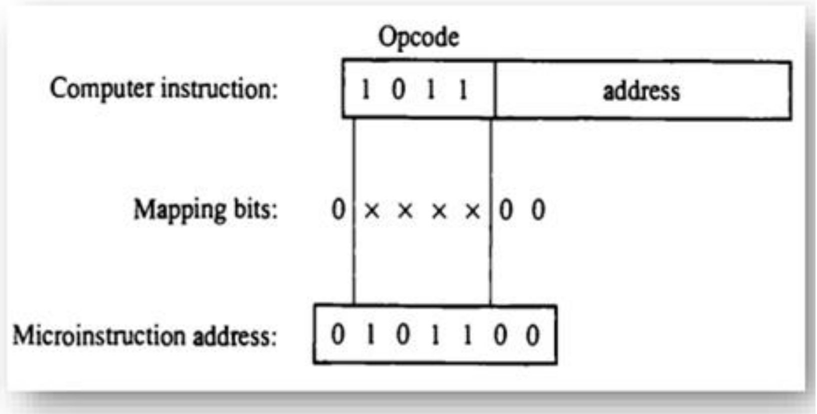
Selection of Address for Control Memory

- **A 1 output in the multiplexer generates a control signal to transfer the branch address from the microinstruction into the control address register.**
- **A 0 output in the multiplexer causes the address register to be incremented.**
- A special type of branch exists when a microinstruction specifies a branch to the first word in control memory where a microprogram routine for an instruction is located.
- The status bits for this type of branch are the bits in the operation code part of the instruction.



Mapping of Instruction

- Mapping of opcode from computer instruction to address of the microinstruction in a control memory.
- For example, a computer with a simple instruction format has an **operation code of four bits** which can specify up to **16 distinct instructions**.
- Assume further that the **control memory has 128 words**, requiring an **address of seven bits**.
- For each operation code there exists a microprogram routine in control memory that executes the instruction.
- **Mapping process that converts the 4-bit operation code to a 7-bit address for control memory.**

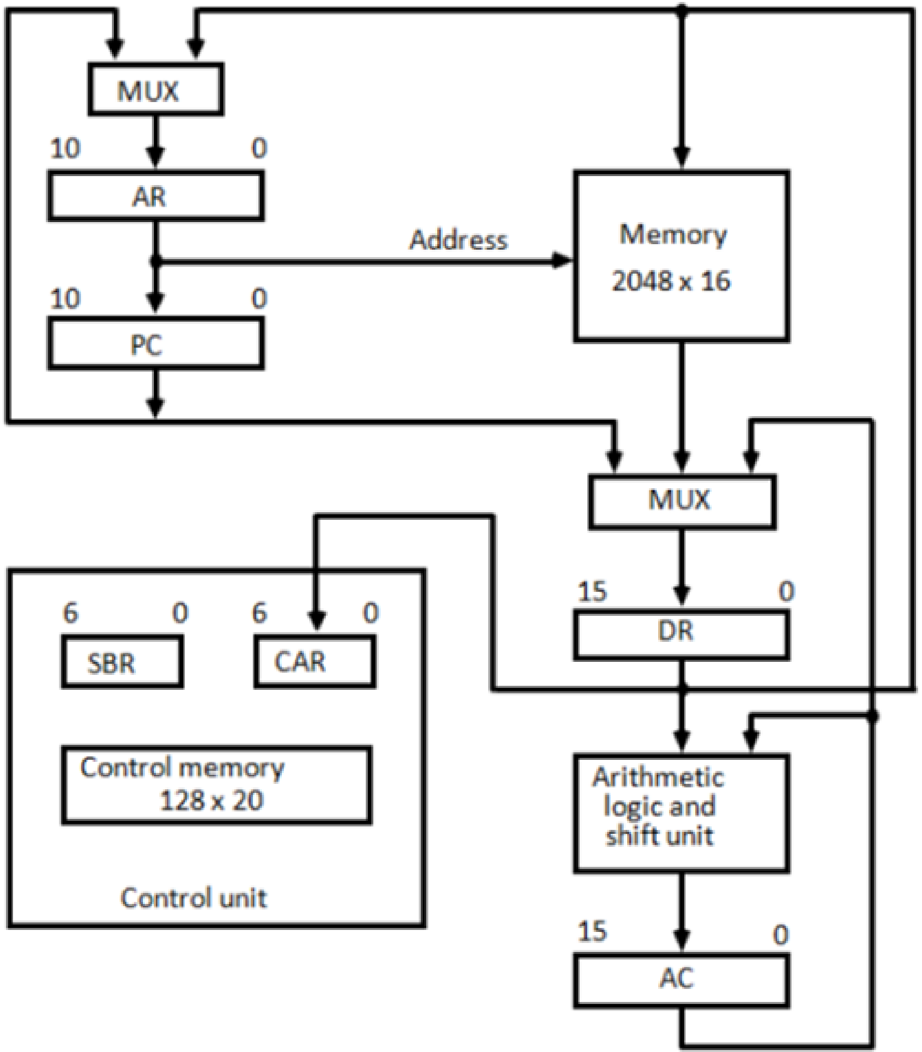


- 3 steps of Mapping:

1. Leave MSB to zero.
2. Leave 2 LSB to zero.
3. Compare address with the rest of digits.

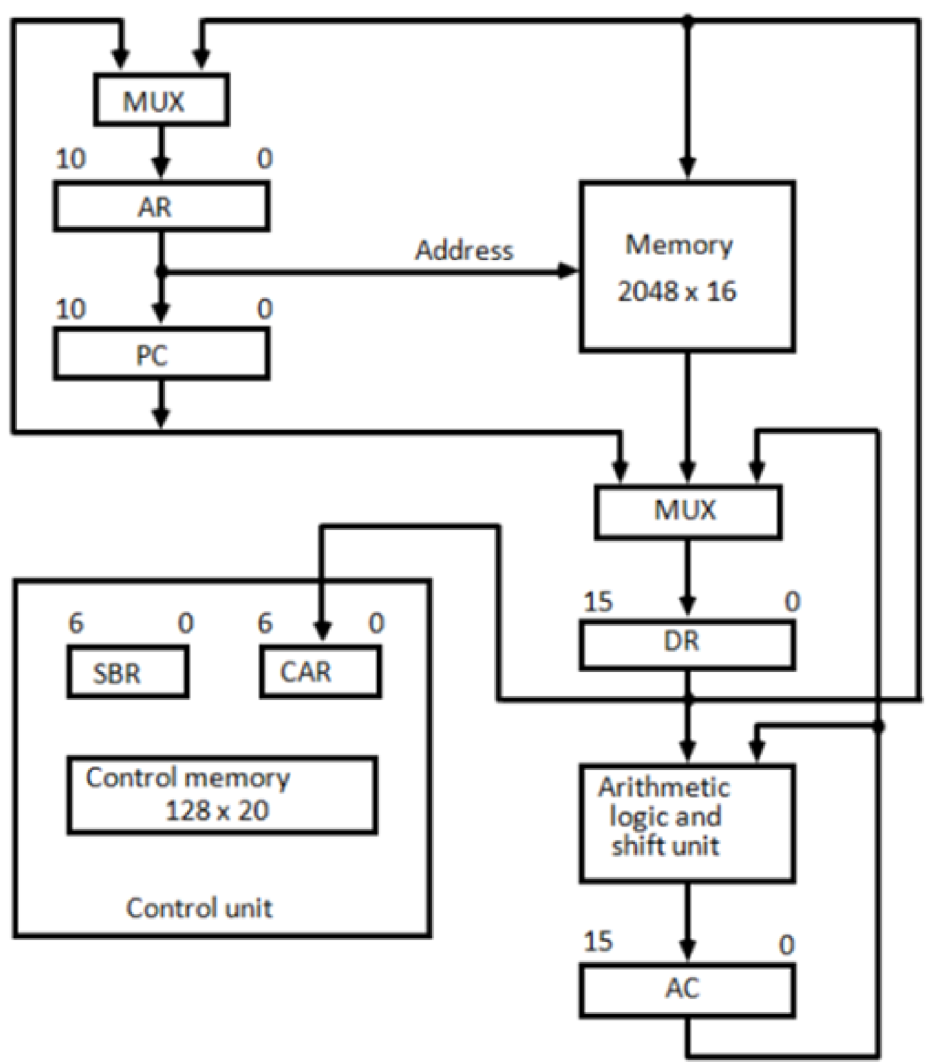
Computer Hardware Configuration (MICROPROGRAM EXAMPLE)

- It consists of **two memory units**: a main memory for storing instructions and data, and a control memory for storing the microprogram.
- **Four registers are associated with the processor unit and two with the control unit.**
- The processor registers are program counter (PC), address register (AR), data register (DR), and accumulator register (AC).
- The control unit has a control address register (CAR) and a subroutine register (SBR).



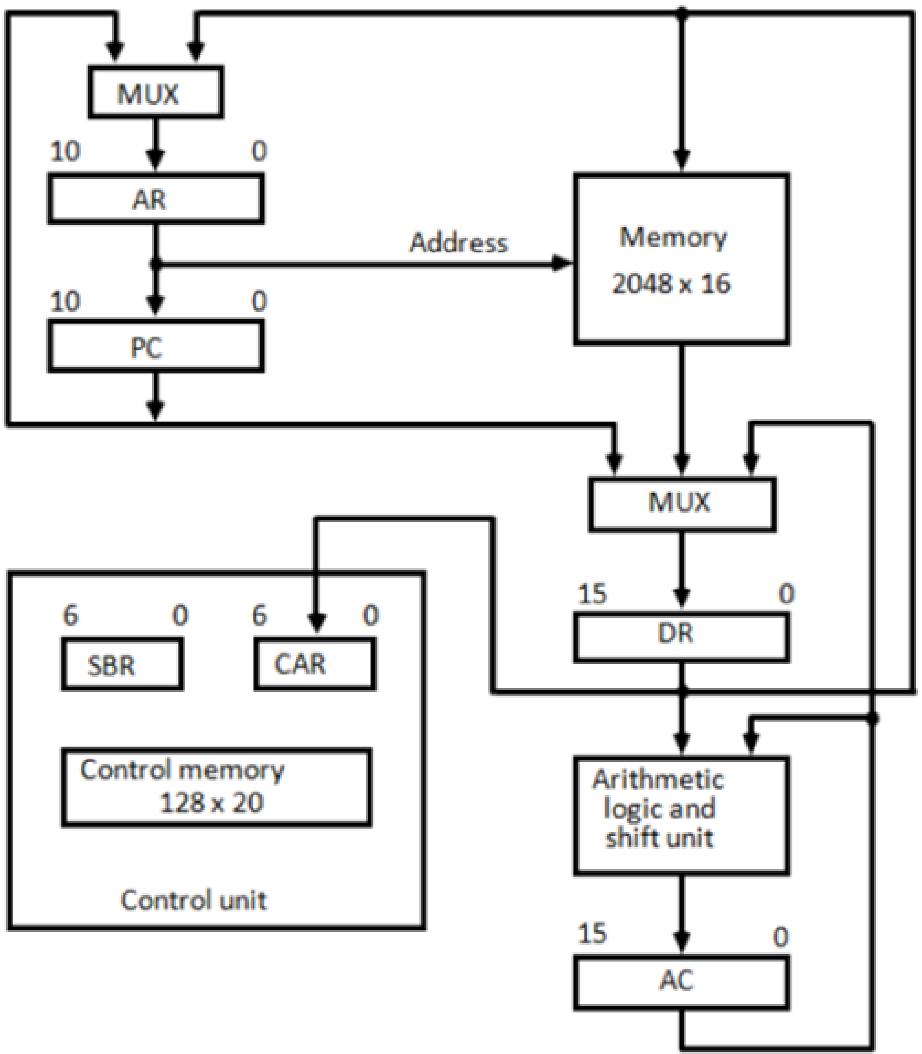
Computer Hardware Configuration (MICROPROGRAM EXAMPLE)

- Size of main memory is of 2048 words and size of each word is 16 bits.
- $2^{11} = 2048$ so that 11 bits are required to specify the address.
- Size of AR=11
- Size of PC=11
- Size of each word = Size of DR = Size of AC = 16 bits
- Size of control memory is 128 words and size of each word is 20 bits.
- $2^7 = 128$ so that 7 bits are required to specify the address.
- Size of word = size of instruction = 20 bits
- So the size of SBR and CAR = 7 bits



Computer Hardware Configuration (MICROPROGRAM EXAMPLE)

- The **transfer of information among the registers** in the processor is done through **multiplexers**.
- DR can receive information from AC, PC, or memory.
- AR can receive information from PC or DR.
- PC can receive information only from AR.
- The arithmetic, logic, and shift unit performs microoperations with data from AC and DR and places the result in AC.



Computer Hardware Configuration
(MICROPROGRAM EXAMPLE)

15	14 – 11	10 – 0
I	Opcode	Address

(a) Instruction format

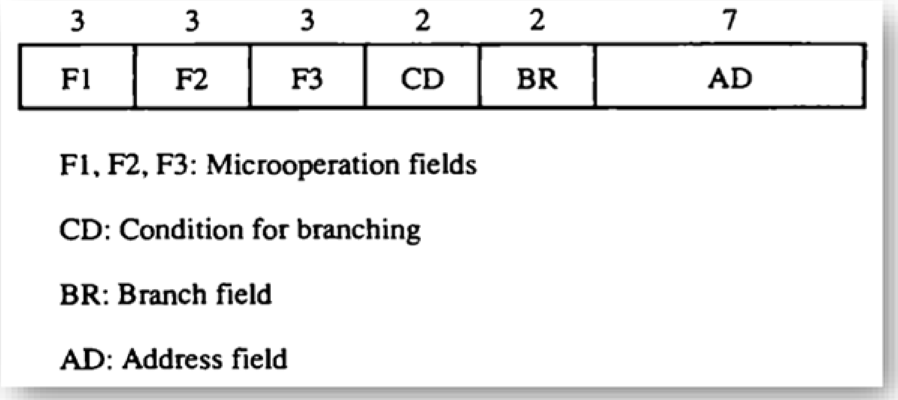
Symbol	Opcode	Description
ADD	0000	$AC \leftarrow AC + M[EA]$
BRANCH	0001	If $(AC < 0)$ then $(PC \leftarrow EA)$
STORE	0010	$M[EA] \leftarrow AC$
EXCHANGE	0011	$AC \leftarrow M[EA], M[EA] \leftarrow AC$

EA is the effective address

(b) Four computer instructions

Microinstruction Format

- The 20 bits(as per the previous size of control memory) of the microinstruction are divided into **four functional parts**.
- The three fields **F1, F2, and F3 specify microoperations (opcode field)** for the computer.



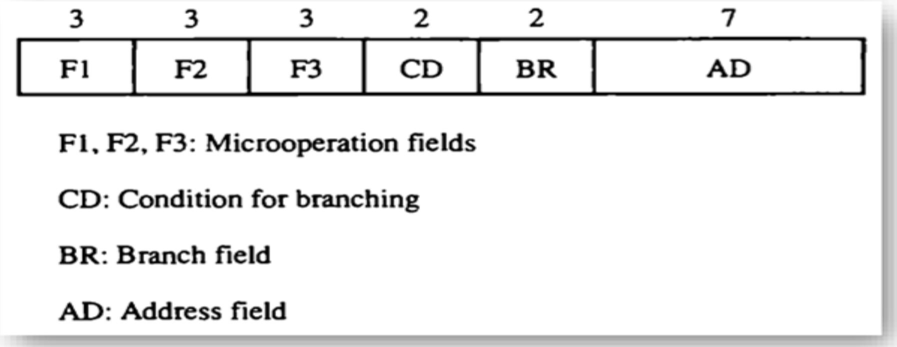
- The CD field specifies **status bit conditions**.
- The BR field specifies the **type of branch** to be used.
- The AD field **contains a address**.

Microinstruction Format

- The microoperation field is subdivided into 3 fields(F1,F2,F3) of three bits each.
- The 3 bits in each fields are encoded to specify seven distinct microoperation plus one idle microoperation(i.e. No Operation). This gives total 21 microoperation.
- **No more than three micro operations can be chosen for a microinstruction**, one from each field. If fewer than three micro operations are used, one or more of the fields will use the binary code 000 for no operation.
- As an illustration, a microinstruction can specify two simultaneous microoperations from F2 and F3 and none from F1.

  DR<-M [AR]   with F2 = 100

  and PC<-PC + 1   with F3 = 101
- The nine bits of the micro operation fields will then be 000 100 101.



Microinstruction Format

F1	Microoperation	Symbol
000	None	NOP
001	$AC \leftarrow AC + DR$	ADD
010	$AC \leftarrow 0$	CLRAC
011	$AC \leftarrow AC + 1$	INCAC
100	$AC \leftarrow DR$	DRTAC
101	$AR \leftarrow DR(0\text{–}10)$	DRTAR
110	$AR \leftarrow PC$	PCTAR
111	$M[AR] \leftarrow DR$	WRITE

F2	Microoperation	Symbol
000	None	NOP
001	$AC \leftarrow AC - DR$	SUB
010	$AC \leftarrow AC \vee DR$	OR
011	$AC \leftarrow AC \wedge DR$	AND
100	$DR \leftarrow M[AR]$	READ
101	$DR \leftarrow AC$	ACTDR
110	$DR \leftarrow DR + 1$	INCDR
111	$DR(0\text{-}10) \leftarrow PC$	PCTDR

F3	Microoperation	Symbol
000	None	NOP
001	$AC \leftarrow AC \oplus DR$	XOR
010	$AC \leftarrow \overline{AC}$	COM
011	$AC \leftarrow \text{shl } AC$	SHL
100	$AC \leftarrow \text{shr } AC$	SHR
101	$PC \leftarrow PC + 1$	INCPC
110	$PC \leftarrow AR$	ARTPC
111	Reserved	

Microinstruction Format

❖ **CD (condition) field:**

- It consists of two bits which are encoded to specify four status bit conditions.
- The first condition is always 1, so that a reference to CD=00 will always find the condition to be true.
- When this condition is used in conjunction with BR field, it provides unconditional branch(U).
- An indirect bit I is available in bit 15 of DR.
- Sign(S) bit of AC is available in bit 15 of AC.
- The zero value(Z) is a binary variable whose value is equal to 1 if all the bits of AC is equal to zero.
- These four symbols we will use as a status bits when we write microprogram in symbolic form.

CD	Condition	Symbol	Comments
00	Always = 1	U	Unconditional branch
01	*DR*(15)	I	Indirect address bit
10	*AC*(15)	S	Sign bit of *AC*
11	*AC* = 0	Z	Zero value in *AC*

❖ BR (branch) field:

- It consists of two bits.
- It is used, in conjunction with the address field AD, to choose the address of the next microinstruction.

BR	Symbol	Function
00	JMP	$CAR \leftarrow AD$ if condition = 1
		$CAR \leftarrow CAR + 1$ if condition = 0
01	CALL	$CAR \leftarrow AD$, $SBR \leftarrow CAR + 1$ if condition = 1
		$CAR \leftarrow CAR + 1$ if condition = 0
10	RET	$CAR \leftarrow SBR$ (Return from subroutine)
11	MAP	$CAR(2\text{–}5) \leftarrow DR(12\text{-}15)$, $CAR(0,1,6) \leftarrow 0$

- When BR = 00, the control performs a jump (JMP) operation (which is similar to a branch), and when BR = 01, it performs a call to subroutine (CALL) operation.
- The two operations are identical except that a call microinstruction stores the return address in the subroutine register SBR. The jump and call operations depend on the value of the CD field.
- The return from subroutine is accomplished with a BR field equal to 10. This causes the transfer of the return address from SBR to CAR .
- The mapping from the operation code bits of the instruction to an address for CAR is accomplished when the BR field is equal to 11.
- **AD (address) field:**
- It contains a branch address.
- The address field is seven bits wide, since the control memory has $128 = 2^7$ words.
- If there is no branch AD field contains address of the next micro instruction.

3	3	3	2	2	7
F1	F2	F3	CD	BR	AD

F1, F2, F3: Microoperation fields

CD: Condition for branching

BR: Branch field

AD: Address field

Example:

ADD	NOP	NOP	U	CALL	Address bits
001	000	000	00	01	0100011

BR	Symbol	Function
00	JMP	$CAR \leftarrow AD$ if condition = 1
		$CAR \leftarrow CAR + 1$ if condition = 0
01	CALL	$CAR \leftarrow AD$, $SBR \leftarrow CAR + 1$ if condition = 1
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CD	Condition	Symbol	Comments
00	Always = 1	U	Unconditional branch
01	$DR(15)$	I	Indirect address bit
10	$AC(15)$	S	Sign bit of AC
11	$AC = 0$	Z	Zero value in AC

GTU Questions

Q.1 State differences between hardwired control unit and microprogrammed control unit.
Q.2 Explain using a flowchart how address of control memory is selected in microprogrammed control unit. OR What is address sequencing? Explain in detail. (4 times) OR Which address sequencing capabilities are required in a control memory?
Q.3 Write the symbolic microprogram routine for the BSA instruction. Use the microinstruction format of basic microprogrammed control unit.
Q.4 Discuss microprogrammed control organization with neat diagram.(2 times)
Q.5 Explain status bit conditions with neat diagram.
Q.6 Define followings: 1. Control Memory 2. Control Word 3. Control Address Register
Q.7 Sketch Microinstruction code format. Quote BR and CD field in brief.
Q.9 Discuss the importance of “control word” in a processor.

GTU Questions

Q.3 Write the symbolic microprogram routine for the BSA instruction. Use the microinstruction format of basic microprogrammed control unit.

$$D_5T_4: \quad M[AR] \leftarrow PC, \quad AR \leftarrow AR + 1$$
$$D_5T_5: \quad PC \leftarrow AR, \quad SC \leftarrow 0$$

Microoperation	Description
PCTDR	DR<-PC
ARTPC	PC<-AR
WRITE	M[AR]<-DR
INCPC	PC<-PC+1

Thanking You